

SAML Assignment 1.2: Hardware Trojan Detection

June 7, 2025

1 Understanding the Data

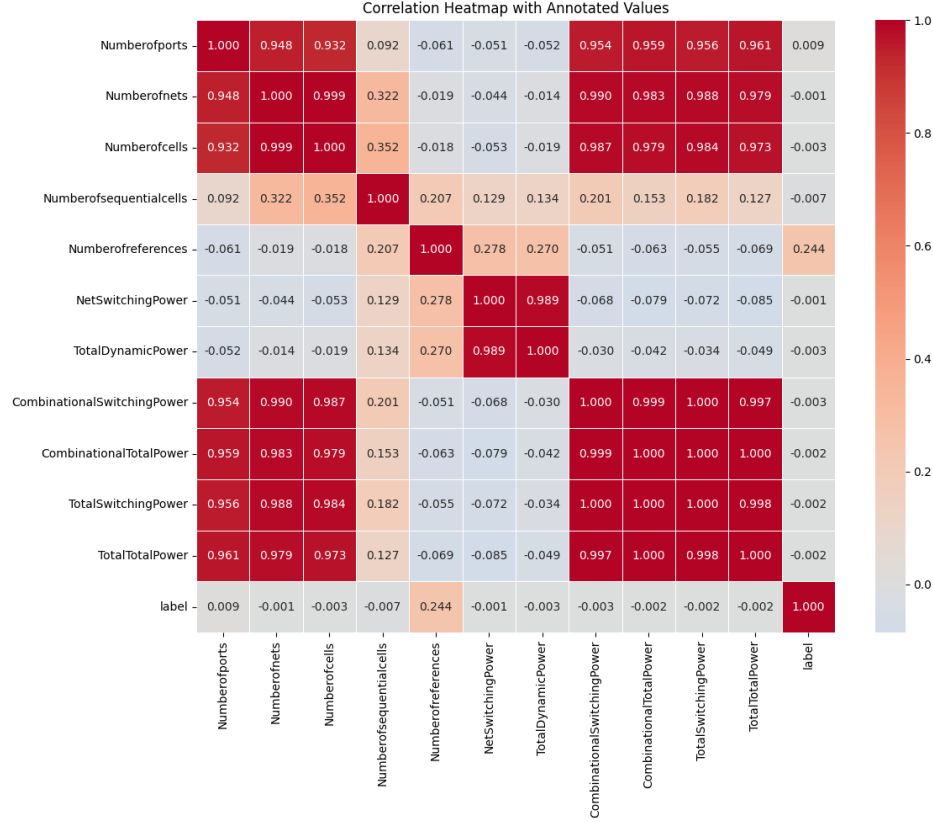
The dataset used in this assignment is a modified version of a public benchmark originally designed to evaluate Hardware Trojan detection in circuits. The provided CSV file contains 1747 entries, with 12 features describing the circuit structure and power characteristics, and a binary label indicating the presence of a Trojan (1 = Trojan, 0 = Benign). Unlike the original dataset, identifying circuit names have been removed to prevent trivial classification.

The dataset includes structural features and power-related features, separated into combinational and sequential contributions. An initial observation revealed that many rows were duplicates: only 879 of the 1747 entries are unique, hinting that Trojan identification might be simplified due to limited variability in circuits.

Basic statistics indicate a high variance and strong skew in several features (e.g., CombinationalTotalPower, TotalTotalPower). Outlier analysis using the interquartile distance method confirmed this: over 200 outliers were found in some power-related features. This supports the hypothesis that malicious circuits introduce noticeable deviations in the circuit structure or power profile.

Outlier counts per feature:	
	# Outliers
Numberofnets	202
TotalTotalPower	202
Numberofcells	198
CombinationalSwitchingPower	196
TotalSwitchingPower	196
CombinationalTotalPower	196
Numberofsequentialcells	194
NetSwitchingPower	140
TotalDynamicPower	138
Numberofports	61
Numberofreferences	0

Correlation analysis shows that no feature is strongly linearly correlated with the target label, although Numberofreferences exhibits slightly higher correlation. This implies that models relying purely on linear relationships may underperform.



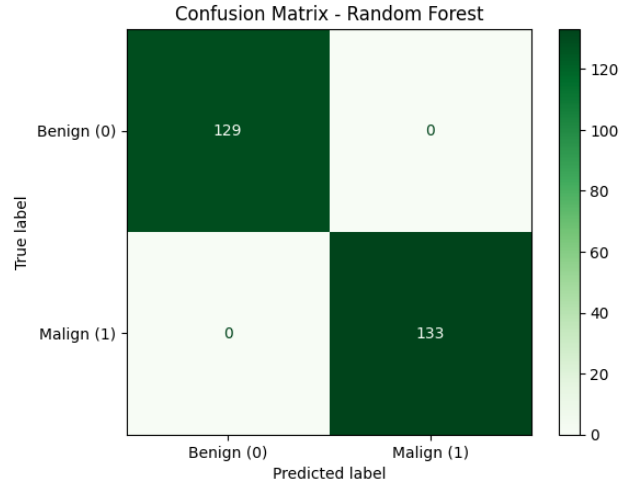
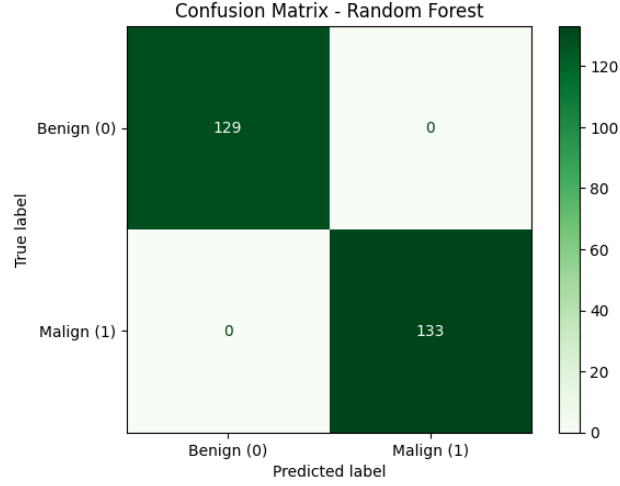
The redundancy of some features (e.g. TotalDynamicPower, TotalSwitchingPower, TotalTotalPower) suggests they may be derived or linearly combined, which might unnecessarily inflate model complexity. This is confirmed by their near perfected correlation.

2 Model Selection and Training

Given the weak correlation patterns and the prior success of in previous (Malware Detection) models, a Random Forest classifier was selected. Random Forests are a nice fit for this task due to their robustness to outliers, ability to handle irrelevant features, and built-in feature importance interpretation. Two models were trained:

1. A full-feature model using all features.
2. A reduced model using only Numberofports, Numberofreferences, and Numberofsequentialcells (all those which correlation is greater than .005 , note those three also have a correlation of less than .10).

Both models achieved perfect classification results on the test set with F1-score = 1.00, precision = 1.00, recall = 1.00, MCC = 1.00. This level of performance is highly unusual and suggests the dataset is trivially separable. Inspection of feature importances confirmed this: in both models, Numberofreferences dominated as the most informative feature.



While these results satisfy the assignment requirement of $F1=0.93$, they raise concerns about generalizability to real-world Trojan detection tasks. As mentioned, the abundance of duplicate rows and narrow benign variability likely simplifies the classification task.

3 Cell Type Analysis

Sequential cells are more complex and can be altered to introduce or trigger malicious behavior with less footprint. My hypothesis is that enabling a wider spectrum of malicious behaviors and requiring modification of less cells would make the sequential cells a prime targets over their simpler counterpart.

To further test this, one could show that two models trained using Sequential or Combinational features and compare the result, however, the simplicity of the task and the unusual dataset (e.g. a lot of outliers, duplicates, easily separable classes) might flaw the result. After trying, it was clear than both model where going to perform in similar near-perfection on the dataset anyway.

The simpler approach I then used was to check the feature importance in both the full and reduced datasets, where Sequential features are shown to affect more the prediction.

Numberofreferences	0.298953
Numberofsequentialcells	0.096467
Numberofnets	0.094085
CombinationalSwitchingPower	0.075189
CombinationalTotalPower	0.074856
TotalDynamicPower	0.072155
TotalSwitchingPower	0.069383
NetSwitchingPower	0.066179
TotalTotalPower	0.065313
Numberofcells	0.064819

Numberofreferences	0.519783
Numberofsequentialcells	0.332245
Numberofports	0.147972

4 Conclusion

This assignment explored the application of Random Forests to the problem of hardware Trojan detection using circuit-level features. Despite achieving perfect predictive performance on both full and reduced models, further inspection revealed the dataset likely favors such outcomes due to structural simplicity and limited variability. The analysis also provides some evidences that features related to Sequential cells provide more discriminatory power.